LESSON 1.2

Math is Exploring and Thinking

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make sense of a problem, plan a strategy, and use fractions of a whole number to compare quantities.

WHY THIS MATTERS Doing math is exploring: you make sense of a problem, plan a strategy, check your progress, and try new paths when you hit a dead end. Today's problems — walking distances and skyscraper heights — reward thinkers, not just calculators. Try this one: a new building will be 3/4 as tall as a tower that is 400 meters tall. Plan a strategy to find how tall the new building will be.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
quantity *Spanish: cantidad*	An amount in a problem — a number with a meaning attached, like 540 meters.	540 meters — the number 540 AND what it measures.
relationship *Spanish: relación*	How two quantities compare or connect to each other.	1 hour → 48 km, 2 hours → 96 km: as one grows, so does the other.
strategy *Spanish: estrategia*	A plan for how to solve a problem before you start computing.	"I will draw a tape diagram first, then divide" — chosen before computing.
reasonable *Spanish: razonable*	A solution that makes sense in the context of the problem.	5/6 of 540 must be under 540, so 450 is reasonable and 650 is not.
persevere *Spanish: perseverar*	To keep working on a problem, checking your progress and adjusting your plan when you get stuck.	Stuck on the tram problem → try a tape diagram instead → keep going.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bowl is full of peas — far too many to count one at a time. About how many peas are in the bowl?

1. I do not count every pea — I look for a group I CAN count, like one small cluster of about 10.
2. Then I ask how many of those clusters would cover the whole bowl. I count about 12.
3. 12 clusters of about 10 is 12 × 10 = 120, so my estimate is about 120 peas — a reasoned answer, close enough to be useful.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

105.76 = 100 + 5.76.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To know _____, I first need _____, because _____ depends on it.
- I know that _____ .
- I don't know _____ .
- When I'm stuck, I can _____ .

Word bank quantity | relationship | strategy | reasonable | persevere

WATCH OUT!

A common mistake

Expecting a fraction of a number to be bigger than the number

LESSON 1.2

Math is Exploring and Thinking

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Which of these is a correct way to decompose 105.76?

(A) 10 + 5.76
(B) 100 + 5.76
(C) 100 + 5.076
(D) 105 + 7.6

STEPS TO TRY
1. Add each option and compare to 105.76.
2. The parts must total 105.76 exactly.
3. Add the two parts in each choice. Only one total lands exactly on 105.76.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which is another correct decomposition of 105.76?

(A) 106 − 0.34
(B) 110 − 4.24
(C) 110 − 5.24
(D) 110 + 4.24

STEPS TO TRY

1. Start at 110. How far down is 105.76?
2. 110 − 105.76 = 4.24.
3. So subtracting 4.24 from 110 works.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Deon walked a shorter distance than Miguel, and Evelyn walked a longer distance than Miguel. Who walked the shortest distance?

(A) Deon
(B) Miguel
(C) Evelyn
(D) It cannot be determined

STEPS TO TRY

1. Put Miguel in the middle and place the other two around him.
2. Deon is below Miguel; Evelyn is above Miguel.
3. Who ends up lowest?

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

4 **Answer in complete sentences.**

In the Comparing Walks problem, name one thing you KNOW and one thing you DON'T know.

SENTENCE STARTER I know that ___ .
I don't know ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick one card you put under 'We DON'T know this yet.' What would you have to figure out first to move it to the known column, and which quantity does it depend on?

SENTENCE STARTER To know ___, I first need ___, because ___ depends on it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help